

# PROJECT DOCUMENTATION OF LEAVE MANAGEMENT SYSTEM

## Abstract

The Leave Management System is a ServiceNow-based application developed to automate the employee leave request process within an organization. The system allows employees to submit leave requests through a custom application. The application routes requests for manager approval, automatically calculates leave duration, validates user inputs, sends notifications, and provides dashboard reporting for request monitoring.

The project reduces manual effort, improves transparency, enhances communication, and provides a centralized platform for tracking employee leave requests.

## Introduction

Employee leave management is an essential process in every organization. Traditional methods of managing leave requests often involve emails, spreadsheets, or paper-based forms, which may lead to delays, calculation errors, lack of visibility into request status, and inefficient record management.

To overcome these challenges, the Leave Management System was developed using ServiceNow. The system automates leave request submission, approval workflows, leave duration calculation, validation management, notification generation, and reporting.

## Problem Statement

Many organizations still use manual processes to manage employee leave requests.

Challenges include:

- Manual leave request handling
- Approval delays
- Lack of request tracking
- Errors in leave duration calculation
- Increased administrative workload
- Communication gaps
- Difficulty maintaining leave records

These challenges reduce efficiency and increase processing time.

## Objectives

The objectives of the project are:

### Primary Objectives

- Automate employee leave requests.
- Reduce paperwork and manual intervention.

- Improve approval efficiency.
- Enable leave request tracking.
- Automate leave duration calculation.
- Provide dashboard reporting.
- Improve leave management efficiency.

### **Secondary Objectives**

- Improve user experience.
- Ensure process transparency.
- Maintain leave request history.
- Reduce calculation errors.
- Improve communication through notifications.

### **Development Environment**

#### **Platform Used**

ServiceNow Developer Instance

#### **Development Modules**

- ServiceNow App Studio
- Flow Designer
- Business Rules
- Client Scripts
- Custom Tables
- Dashboards and Reports
- Form Designer
- Notifications

### **Application Development**

#### **Step 1: Creation of Application**

A custom application named:

#### **Leave Management System**

was created using ServiceNow App Studio.

#### **Purpose**

The application serves as a centralized platform for managing employee leave requests.

## Application Components

- Custom Table
- Leave Request Form
- Approval Flow
- Business Rule
- Client Script
- Dashboard
- Reports
- Notifications

### 4. Custom Table Development

A custom table was created to store employee leave request information.

#### Table Name

Leave Request

#### Fields Created

| Field Name      | Description               |
|-----------------|---------------------------|
| Employee Name   | Employee requesting leave |
| Employee ID     | Employee identifier       |
| Leave Type      | Type of leave requested   |
| Start Date      | Leave start date          |
| End Date        | Leave end date            |
| Total Days      | Leave duration            |
| Reason          | Purpose of leave          |
| Approval Status | Approval state            |

#### Purpose

The table stores all leave request records and acts as the primary data repository for the application.

### 5. Form Design Development

A user-friendly form was designed for employee leave request submission.

#### Form Fields

### **Employee Information**

- Employee Name
- Employee ID

### **Leave Information**

- Leave Type
- Start Date
- End Date
- Total Days
- Reason

### **Validation**

Mandatory fields were configured to ensure complete leave request submission.

### **Benefits**

- Improved usability
- Reduced data entry errors
- Standardized leave request collection
- Improved data quality

## **6. Leave Request Form Development**

A custom leave request form was created to allow employees to submit leave requests.

### **Form Name**

Leave Request Form

### **Variables Created**

| Variable      | Type             |
|---------------|------------------|
| Employee Name | Single Line Text |
| Employee ID   | Single Line Text |
| Leave Type    | Choice           |
| Start Date    | Date             |
| End Date      | Date             |
| Total Days    | Integer          |
| Reason        | Multi-Line Text  |

| Variable        | Type   |
|-----------------|--------|
| Approval Status | Choice |

### **Functionality**

Employees can submit leave requests using a structured form.

## **7. Workflow Development Using Flow Designer**

A workflow was created using Flow Designer to automate the leave request lifecycle.

### **Flow Name**

Leave Approval Flow

### **Trigger**

Leave Request Created

The workflow starts automatically whenever a new leave request record is created.

### **Approval Action**

An approval step was configured using:

### **Ask For Approval**

### **Purpose**

The request is routed to the reporting manager for review and approval.

### **Approval Rule**

Anyone Approves

### **Conditional Logic**

An IF condition was configured.

### **Condition**

Approval State = Approved

### **Purpose**

The workflow proceeds only when approval is granted.

### **Status Management**

The request status is automatically updated based on the manager's decision.

Approved → Status = Approved

Rejected → Status = Rejected

### **Notification Management**

Employees receive notifications when:

- Leave Request Submitted
- Leave Request Approved
- Leave Request Rejected

## **8. Business Rule Development**

A Business Rule was configured to automatically calculate the total number of leave days.

### **Business Rule Name**

Calculate Total Leave Days

### **Purpose**

Automatically calculates leave duration based on the selected Start Date and End Date.

### **Example**

Start Date : 23-Jun-2026

End Date : 25-Jun-2026

Total Days : 3

### **Benefits**

- Eliminates manual calculations
- Improves accuracy
- Reduces processing time
- Ensures consistent leave calculations

## **9. Client Script Development**

A Client Script was configured to validate leave dates before submission.

### **Client Script Name**

Leave Date Validation

### **Validation Rule**

End Date cannot be earlier than Start Date.

### **Error Message**

"End Date cannot be earlier than Start Date"

### **Purpose**

Prevents invalid leave requests and improves data quality.

## **10. Dashboard Development**

A dashboard was created to monitor leave requests and approval status.

## **Dashboard Components**

- Total Requests
- Approved Requests
- Rejected Requests
- Pending Requests
- Monthly Leave Statistics

## **Purpose**

Provides real-time visibility into leave request activities and approval trends.

## **11. Reporting Features**

Reports were created to analyze leave request data.

### **Report Types**

- Total Leave Requests Report
- Approved Requests Report
- Rejected Requests Report
- Pending Requests Report
- Monthly Leave Analysis Report

### **Benefits**

- Improved monitoring
- Better decision making
- Enhanced reporting capabilities

### **Results**

The project successfully automated the employee leave request process.

### **Achievements**

- Automated leave request submission
- Automated approval workflow
- Automatic leave duration calculation
- Client-side validation
- Dashboard-based monitoring
- Centralized leave management
- Automated notifications

## **Benefits**

- Reduced processing time
- Increased transparency
- Improved efficiency
- Better tracking and reporting
- Improved data accuracy
- Enhanced employee experience

## **Conclusion**

The Leave Management System successfully automates the complete employee leave request lifecycle using ServiceNow. The application provides a structured workflow for leave request submission, approval management, automatic leave calculation, validation, notification management, and dashboard reporting.

By reducing manual effort, improving process visibility, and enhancing communication, the system increases organizational efficiency, improves record management, and provides a reliable and user-friendly leave management solution.